

PROJECT DOCUMENTATION

Tesla Inc.

Integrated Financial Model

& Valuation

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DATE	July 2026
SOFTWARE USED	Microsoft Excel

\$24.23

DCF VALUE / SHARE

\$408.95

MARKET PRICE

10.41%

WACC

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Project Overview

Objective: Estimate the intrinsic value of Tesla, Inc. common stock using a bottom-up, driver-based discounted cash flow methodology, benchmarked against the market price.

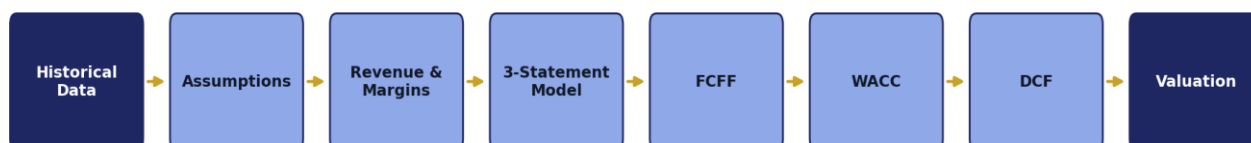
Model Scope: Automotive (5 vehicle lines: Model 3, Model S/X, Model Y, Cybertruck, Semi) and Energy & Other segments. Explicit forecast FY2026–FY2032, built on 5 years of historical data (FY2021–FY2025) from Tesla's SEC filings.

Valuation Methods: Free Cash Flow to Firm (FCFF), a bottom-up Hamada beta and CAPM-derived WACC, and a Gordon Growth terminal value - following the general approach to intrinsic valuation described in Aswath Damodaran's corporate finance and valuation work, Adapted to available data.

Key Features

- Single scenario toggle drives Best/Base/Worst case across every operating driver simultaneously
- Integrated three-statement model with built-in balance and cash reconciliation checks
- Bottom-up cost of capital: beta unlevered/relevered at a target (not current) capital structure
- Two-way sensitivity analysis on WACC and terminal growth, built with live formulas.

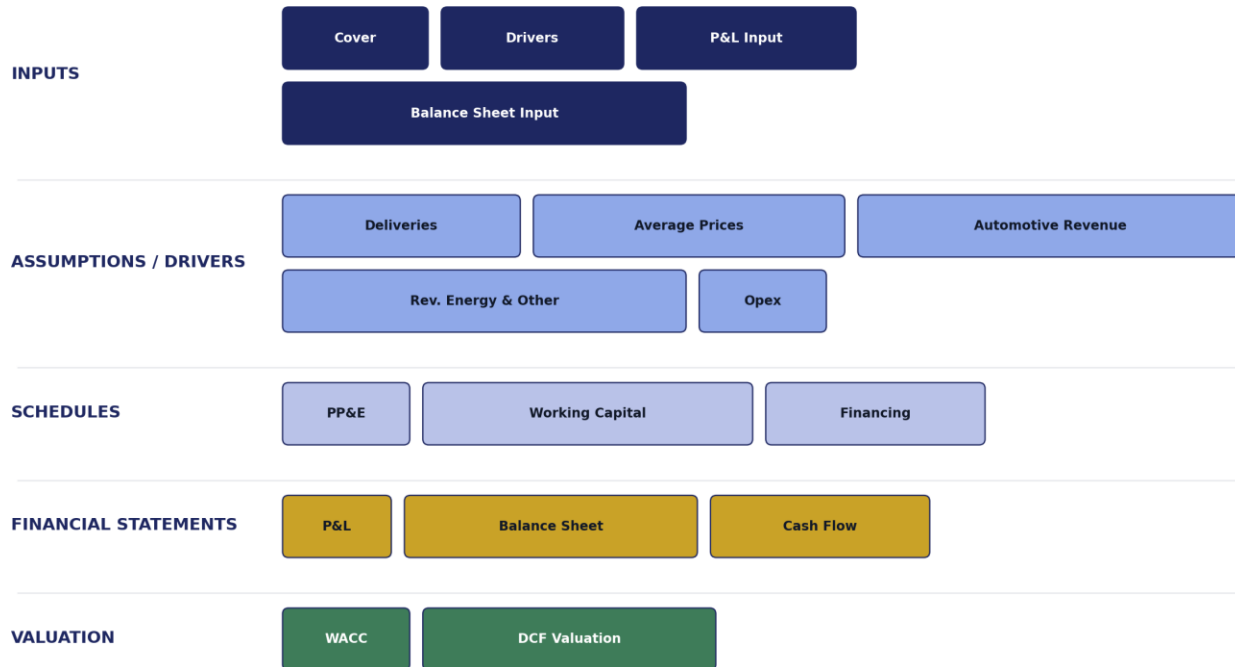
Model Flow



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Workbook Structure

The workbook's 38 sheets are organized into five functional groups, shown below with representative tabs from each.



Group	Contains
Inputs	Cover, Drivers (global assumptions), historical P&L / Balance Sheet data from 10-K filings
Assumptions / Drivers	Deliveries, pricing, and revenue build-up by vehicle line and segment
Schedules	PP&E (capex/depreciation), Working Capital (DSO/DIO/DPO), Financing (debt/equity)
Financial Statements	Fully linked P&L, Balance Sheet, and Cash Flow
Valuation	WACC and DCF Valuation, including the sensitivity table

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Assumptions & Forecast Methodology

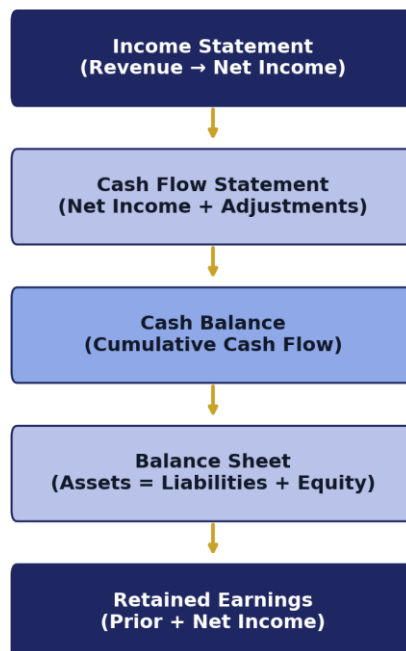
Assumption	Value	Why
Revenue growth	Deliveries × ASP, by line	Bottom-up build reflects product-mix shifts
Gross margin	16.4–18.3% (fcst.)	Anchored to Tesla's own historical average, cross-checked to peers
Capex	% of prior-yr PP&E	Ties spending to asset base
Depreciation	10-yr SL, vintage-based	Each capex tranche depreciates on its own schedule
Working capital	5-yr avg DSO/DIO/DPO	Smooths year-to-year noise; peer-benchmarked
Tax rate	21.0% (marginal)	US marginal tax rate
WACC	10.41%	CAPM cost of equity + after-tax cost of debt, at target capital structure
Terminal growth	4.55% (= risk-free rate)	

A single scenario toggle (`Drivers!C3`) switches every driver above between Best/Base/Worst simultaneously.

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Financial Model Design

The income statement, balance sheet, and cash flow statement are fully linked: net income flows into retained earnings and the cash flow statement's closing cash balance flows into the balance sheet and the balance sheet's ending balances (receivables, inventory, payables, PP&E, debt) feed back into next period's schedules.



Supporting Schedules: -

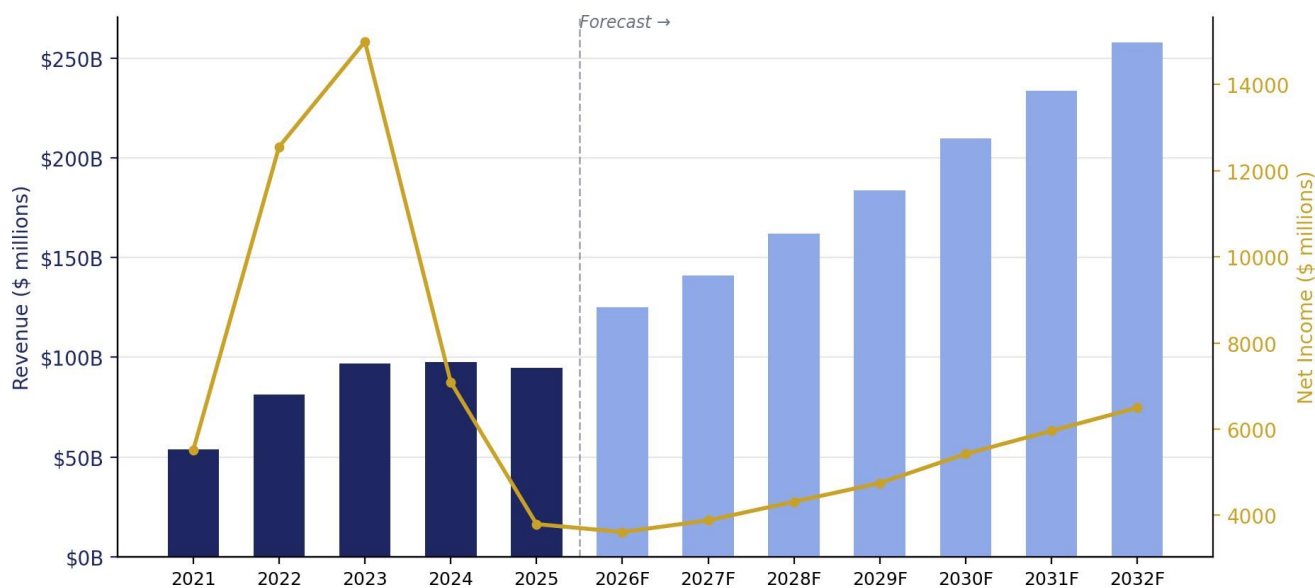
- **PP&E** - vintage-based capex/depreciation schedule feeding the balance sheet and D&A add-back in FCFF
- **Working Capital** - DSO/DIO/DPO-driven receivables, inventory, and payables
- **Financing** - 50/50 debt-equity plug for any period of negative free cash flow

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Financial Projections

Revenue grows from \$94.8B (FY2025 actual) to \$257.9B by FY2032 under the Base Case, a 15.4% CAGR - driven by continued Model Y/3 volume, Cybertruck and Semi ramp, and Energy segment expansion.

Revenue and Net Income: Historical & Forecast (Base Case)



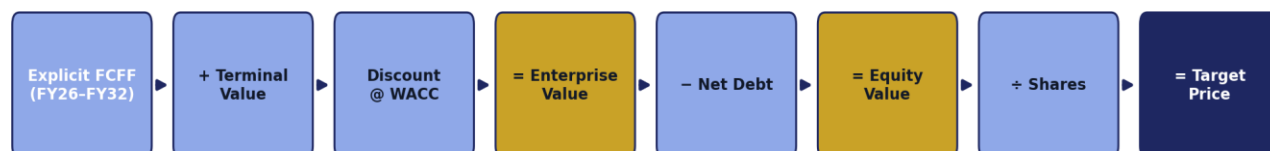
\$ millions	2023A	2025A	2026F	2029F	2032F
Revenue	96,773	94,827	125,044	183,856	257,936
EBIT	8,891	4,355	4,937	6,394	8,589
Net Income	14,997	3,794	3,610	4,747	6,495
EBIT Margin %	9.2%	4.6%	3.9%	3.5%	3.3%

EBIT margin compresses from 9.2% (FY2023) to ~3.3% by FY2032, as operating expense assumptions converge toward a peer-benchmarked average rather than continuing Tesla's own historical trend.

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Valuation Methodology

FCFF = NOPAT + D&A; – Capex – Δ Non-Cash Working Capital. **WACC** blends CAPM cost of equity and after-tax cost of debt at the target capital structure. The **DCF** discounts explicit-period FCFF and a Gordon Growth terminal value back to the present at WACC, using mid-year convention.



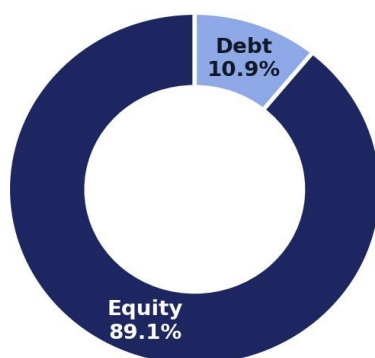
Output	Base Case
Enterprise Value	\$82,800M
Value of Equity	\$90,937M
Value per Share	\$24.23
Current Market Price	\$408.95

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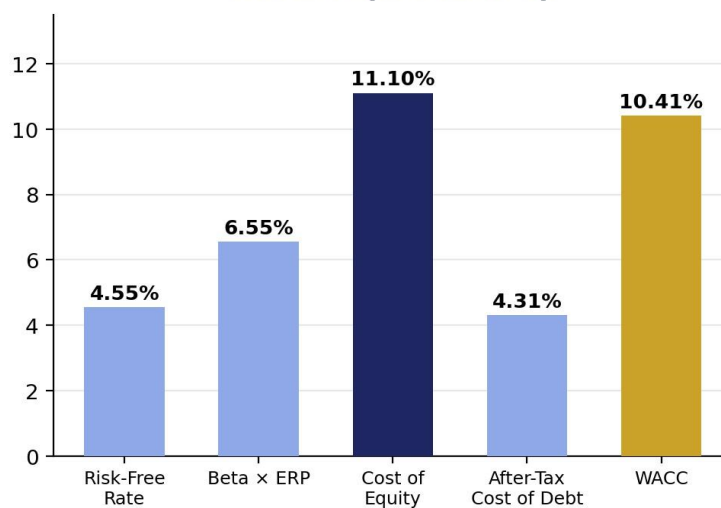
Cost of Capital (WACC)

Beta is unlevered year-by-year using Tesla's own historical D/E ratios (Hamada approach), then relevered at a target capital structure — the 5-year average debt weight, not current-period leverage.

Target Capital Structure

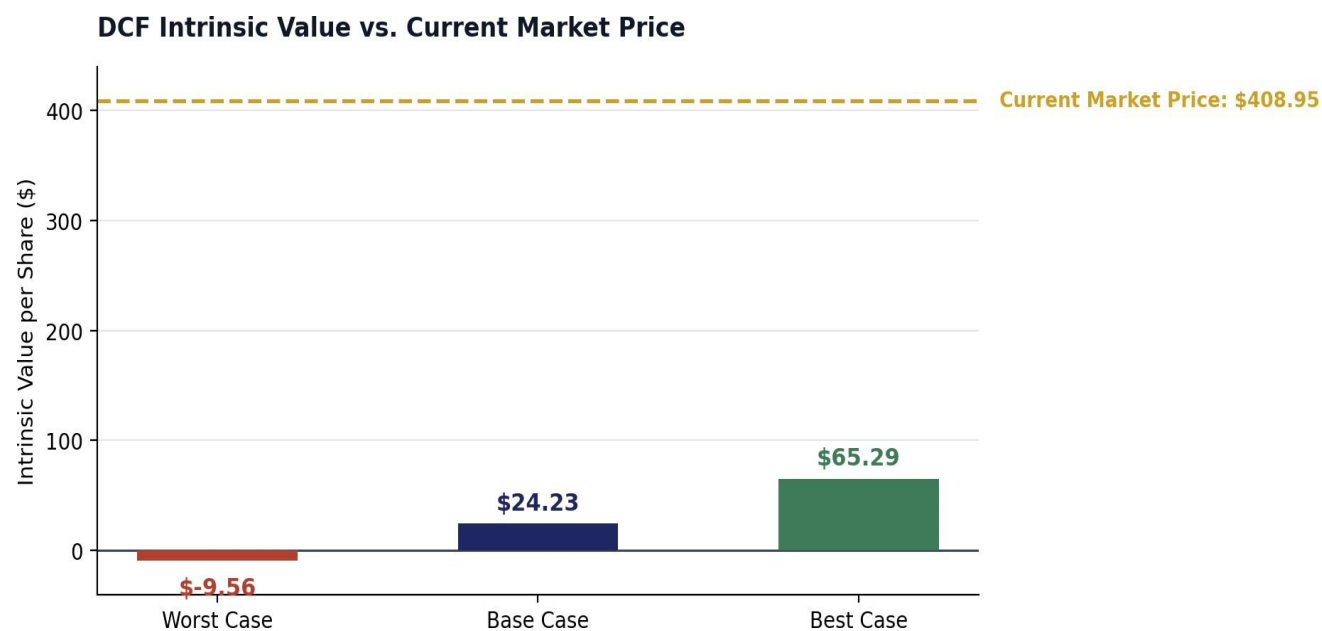


Cost of Capital Build-Up



Component	Value
Cost of Equity (CAPM)	11.10%
After-tax Cost of Debt	4.31%
Target Equity / Debt Weight	89.1% / 10.9%
WACC	10.41%

Key Outputs — Charts & Analysis



Key Findings

- Against the base Case intrinsic value (\$24.23/share) the market is pricing Tesla at 16.9x this DCF's estimate of core-business fundamental value
- The gap is explained by DCF scope - this model excludes FSD, Robotaxi, and Optimus optionality the market may be pricing separately
- Scenario range spans -\$9.56 (Worst) to \$65.29 (Best), reflecting high sensitivity to delivery volume and margin assumptions
- Terminal value highly sensitive to the WACC-growth spread

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Model Checks

Practice	Status
Balance sheet balances (Assets = Liab. + Equity)	✓ Zero variance, every period
Cash flow reconciles to balance sheet cash	✓ Zero variance, every period
No formula errors (#REF!, #VALUE!, #DIV/0!)	✓ 0 errors across 3,357 formulas
No circular references	✓ Confirmed
No hardcoded values inside calculation formulas	✓ Inputs isolated to Drivers/comparable tabs
Consistent color coding (input / formula / link)	✓ Applied throughout
Assumptions separated from calculations	✓ Single Drivers tab controls all scenarios
Dynamic formulas	✓ Sensitivity table uses live formulas, not a Data Table

Limitations & Risks

- **Forecast uncertainty** - Scenario bands vary by driver type (relative % for volumes, fixed points for margins/opex), so the Best/Worst cases are not based on a single uniform adjustment.
- **Assumption dependence** - Operating expense and working capital peer benchmarks use 2019 data and therefore have a recency limitation.
- **DCF sensitivity** - Terminal value -the majority of total value - is highly sensitive to the WACC-minus-growth spread.
- **Macroeconomic risk** - Risk-free rate, terminal growth, and inflation are point-in-time market inputs that will change as conditions change.
- **Company-specific risk** - Excludes FSD licensing, robotaxi, Optimus, and energy storage optionality, which may contribute to the difference between the model value and market price.

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Conclusion

This project includes a fully integrated, auditable three-statement financial model and applies a bottom-up DCF valuation to a publicly traded company, from raw 10-K data through to an intrinsic value estimate.

Skills Demonstrated

Financial modeling · corporate finance · valuation · forecasting · accounting · Excel · scenario analysis · capital budgeting · financial statement analysis

Learning Outcomes

Applying Damodaran-consistent cost of capital methodology in practice wherever possible with limitation to available data; the discipline of building self-checking financial statements and clearly identifying the limitations of the valuation.

Future Improvements

- Scenario manager beyond the current single-toggle approach
- Monte Carlo simulation on key operating drivers
- Relative valuation (comparable company multiples) and a football field summary
- Power BI dashboard layer

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Formula Dictionary

Term	Formula
FCFF	$\text{NOPAT} + \text{D\&A} - \text{Capex} - \Delta \text{Non-Cash Working Capital}$
NOPAT	$\text{EBIT} \times (1 - \text{Marginal Tax Rate})$
WACC	$(E/V \times \text{Cost of Equity}) + (D/V \times \text{Cost of Debt} \times (1 - \text{Tax Rate}))$
Cost of Equity (CAPM)	$\text{Risk-Free Rate} + \text{Beta} \times \text{Equity Risk Premium}$
Levered Beta (Hamada)	$\text{Unlevered Beta} \times (1 + (1 - \text{Tax Rate}) \times D/E)$
Terminal Value	$\text{FCFF}(n+1) \div (\text{WACC} - \text{Terminal Growth Rate})$
ROIC	$\text{NOPAT} \div \text{Invested Capital}$
Reinvestment Rate	$(\text{Net Capex} + \Delta \text{Non-Cash Working Capital}) \div \text{NOPAT}$
Fundamental Growth Rate	$\text{ROIC} \times \text{Reinvestment Rate}$
Working Capital	$\text{Current Assets (ex-cash)} - \text{Current Liabilities (ex-debt)}$
EBITDA	$\text{EBIT} + \text{Depreciation \& Amortization}$
Enterprise Value	$\text{PV of Explicit FCFF} + \text{PV of Terminal Value}$
Equity Value	$\text{Enterprise Value} - \text{Net Debt} - \text{Minority Interests}$
Value per Share	$\text{Equity Value} \div \text{Diluted Shares Outstanding}$

References

- Tesla, Inc. Annual Reports (Form 10-K), FY2021–FY2025 — U.S. Securities and Exchange Commission, sec.gov/edgar
- Tesla, Inc. Quarterly Delivery and Production Reports — ir.tesla.com
- Aswath Damodaran, Corporate Finance and Valuation datasets — pages.stern.nyu.edu/~adamodar
- Tesla, Inc. pricing and vehicle specifications — tesla.com
- U.S. Department of the Treasury — 10-Year Treasury Constant Maturity Rate (risk-free rate)
- Peer company financial data — respective SEC filings (GM, Ford, Stellantis/Fiat Chrysler, BMW, Mercedes-Benz Group, Volkswagen, Scania, MAN, PACCAR)

Disclaimer

This document was prepared as part of an independent financial modeling and valuation portfolio project. It does not constitute investment advice, a research report from a registered investment adviser or broker-dealer, or a recommendation to buy, sell, or hold any security. Historical financial data is sourced from Tesla, Inc.'s public SEC filings; forward-looking figures are model estimates only.